



Thermal performance evaluation of electronic fluorinated liquids for single-phase immersion liquid cooling

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ABSTRACT

The regulation and prediction of the cooling performance by thermal properties of electronic fluorinated liquids (EFLs) are critical challenges in EFL selection and development for single-phase immersion liquid cooling (SPILC) systems in data centers. To address this, this paper proposes a figure of merit (FOM) to guide the selection and development of EFLs for SPILC systems. An integrated experimental platform is constructed to measure EFL thermophysical properties over a wide temperature range. The influence mechanism of thermophysical properties of different EFLs on the flow and heat transfer performance of SPILC systems is numerically investigated. The results indicate that FOM is effective and reliable in evaluating EFL performance. According to the importance of FOM, the ranking order of thermophysical properties for EFLs is dynamic viscosity > thermal conductivity > density > specific heat capacity. With increasing temperature, only the weight factor of dynamic viscosity increases, while weight factors of other thermophysical properties decrease. Moreover, the average Nusselt number and pressure drop of a SPILC cabinet using the EFL with the lowest dynamic viscosity increases by 57.3% and decreases by 59%, respectively. Therefore, when selecting and designing EFLs in data centers, efforts should be made to minimize their dynamic viscosity.

1. Introduction

It is well known that with the rise of emerging industries such as big data and cloud computing, the trend towards high-density and integrated data centers has led to increasingly serious problems with cooling energy consumption [1,2]. In this context, improving cooling technologies to reduce the power usage effectiveness (PUE) of data centers or utilizing latent heat storage technology [3,4] for waste heat recovery has gradually become a key approach to achieving carbon neutrality and carbon peak strategies. Compared to heat pipe [5] and microchannel cooling [6,7], the single-phase immersion liquid cooling (SPILC) technology is expected to be the most promising cooling solution for a new generation of green, low-consumption, and efficient data centers since its high heat-transport efficiency, uniform heat transmission, no risk of leakage, and significantly increased power density.

Several exploratory works have been conducted in SPILC sectors,

focusing mainly on the flow types [8,9], liquid-cooled structures [10,11], and liquid-cooled media [12,13] of SPILC systems. Depending on the flow pattern, SPILC systems in data centers can be divided into buoyancy-driven SPILC systems [14,15], forced convection-driven SPILC systems [16,17], and jet-driven SPILC systems [18,19]. Available studies have demonstrated improved heat transfer performance and energy efficiency in data centers using SPILC systems. Moreover, the SPILC systems differ significantly from conventional air-cooled systems in terms of their liquid-cooled structure [20–22]. Given this, scholars have emphasized the research on the influence mechanism of import and export positions, hotspot distribution characteristics, and flow guide structure on SPILC performance, indicating the development direction of its heat transfer enhancement. In addition, as the refrigerant is in direct contact with high-power electronic devices, the immersion coolants [23–25] have a more significant impact on SPILC performance than flow types and liquid-cooled structures. Thus, research on immersion coolants has been a top priority in the promotion of SPILC data center

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Nomenclature		x_i	coordinate axis
C		<i>Greek symbols</i>	
C	constant	α_k, α_e	model constant
c_p	specific heat capacity	$\Delta\tau$	time variation
C_μ	model constant	Δp	flow resistance loss
C_{1e}, C_{2e}, C_{3e}	model constant	ΔT	temperature argument
$E_{\min}$	the minimum value of the parameter	τ	time
F	function relationship	κ	temperature resistance coefficient
g	gravitational acceleration	λ	thermal conductivity
G_b	generation of turbulence kinetic energy due to buoyancy	ρ	density
G_k	generation of turbulence kinetic energy due to the mean velocity gradients	μ	dynamic viscosity
h	convective heat transport coefficient	μ_{eff}	effective kinetic viscosity coefficient
I	electricity current	δ	the relative uncertainty of measured parameters
k	turbulent kinetic energy	σ_T	Temperature homogeneity coefficient
L	effective length of platinum wire	ε	turbulent kinetic energy dissipation rate
m	mass	η_0	model constant
M	the force-sensitive sensor reading	γ	model constant
N	number of grid nodes	<i>Abbreviation</i>	
Nu	Nusselt number	CPU	central processing unit
$\bar{P}$	mean pressure	DC	direct current
Pr	Prandtl number	DIMM	dual inline memory module
q	heat flux	EFL	electronic fluorinated liquid
R_e	correction factor of the ε equation	FOM	figure of merit
Re	Reynolds number	GPU	graphics processing unit
T	temperature	HDD	hard disk drive
$\bar{T}$	mean temperature	PCB	printed circuit board
T'	temperature pulsation	PCH	platform controller hub
U	voltage	PSU	power supply unit
$\bar{u}_i$	mean velocity component	PUE	Power usage effectiveness
$\overline{u'_i}$	mean velocity pulsation	SPILC	single-phase immersion liquid cooling
V_f	the glass float volume	TEC	thermoelectric cooling
W_f	weight factor		

applications.

Through a systematic literature review, it is confirmed that available investigations on coolants in SPILC systems are focused on compatibility testing [26,27], cooling capacity comparison [22,28], and heat transfer capacity enhancement [29]. However, relatively few studies have been performed regarding the influence mechanism of coolant thermal properties on the SPILC performance. In particular, further studies are necessary on how the thermophysical properties of coolants affect the dynamic heat transport capacities, flow resistance performance, and energy efficiency levels of SPILC data centers. Apart from this, research on evaluation indicators for the performance of coolants in SPILC systems is still in its infancy. Saylor et al. [30] first proposed a figure of merit (FOM) for assessing and comparing the performance of some candidate immersion coolants and qualitatively indicated the relationship between the pump-driven SPILC performance and thermophysical properties such as density, specific heat capacity, and viscosity of coolants. Notably, most existing studies on the FOM of immersion coolants focused on two-phase immersion cooling systems rather than SPILC systems [31,32]. Moreover, existing investigations have yet to establish a quantitative mapping between the thermophysical properties of immersion coolants and the performance of SPILC systems, and the importance of thermophysical properties of immersion coolants on the SPILC performance has yet to be clarified.

To address these issues, this paper undertakes a systematic study of the SPILC performance for data centers, focusing on the role of the thermophysical properties of immersion coolants. The current work designs an experimental platform to measure immersion coolants' thermophysical properties at different temperatures. On this basis, this

paper derives an expression for the FOM that characterizes the immersion coolant performance of SPILC systems, and analyses the extent to which each thermal property of immersion coolants contributes to their overall heat transfer performance. In addition, a numerical investigation of the flow and heat transport properties of SPILC systems with different immersion coolants is performed to demonstrate the generalizability of the FOM. Significantly, the flow and thermal transport characteristics of SPILC cabinets with different immersion coolants are analyzed in depth to reveal the regulation mechanism of thermophysical properties of immersion coolants on the immersion cooling performance, thereby providing a theoretical basis and data support for the selection and development of immersion coolants in practical engineering.

2. Measurement of thermophysical properties

2.1. Experimental setup description

For SPILC data centers, the immersion coolant is usually the electronic fluorinated liquid (EFL). To research the impact of EFLs on SPILC performance, it is necessary to understand the variation of EFL thermophysical properties with temperature. To address this issue, an integrated experimental platform for measuring EFL thermophysical properties is designed and built herein. As presented in Fig. 1(a), the experimental platform includes individual measuring units for thermal conductivity, kinetic viscosity, expansion coefficient, and specific heat capacity. Among these, the temperature-controlled reservoir unit is a significant component of the measurement platform and is mainly used to regulate the EFL temperature. The core element for the temperature-

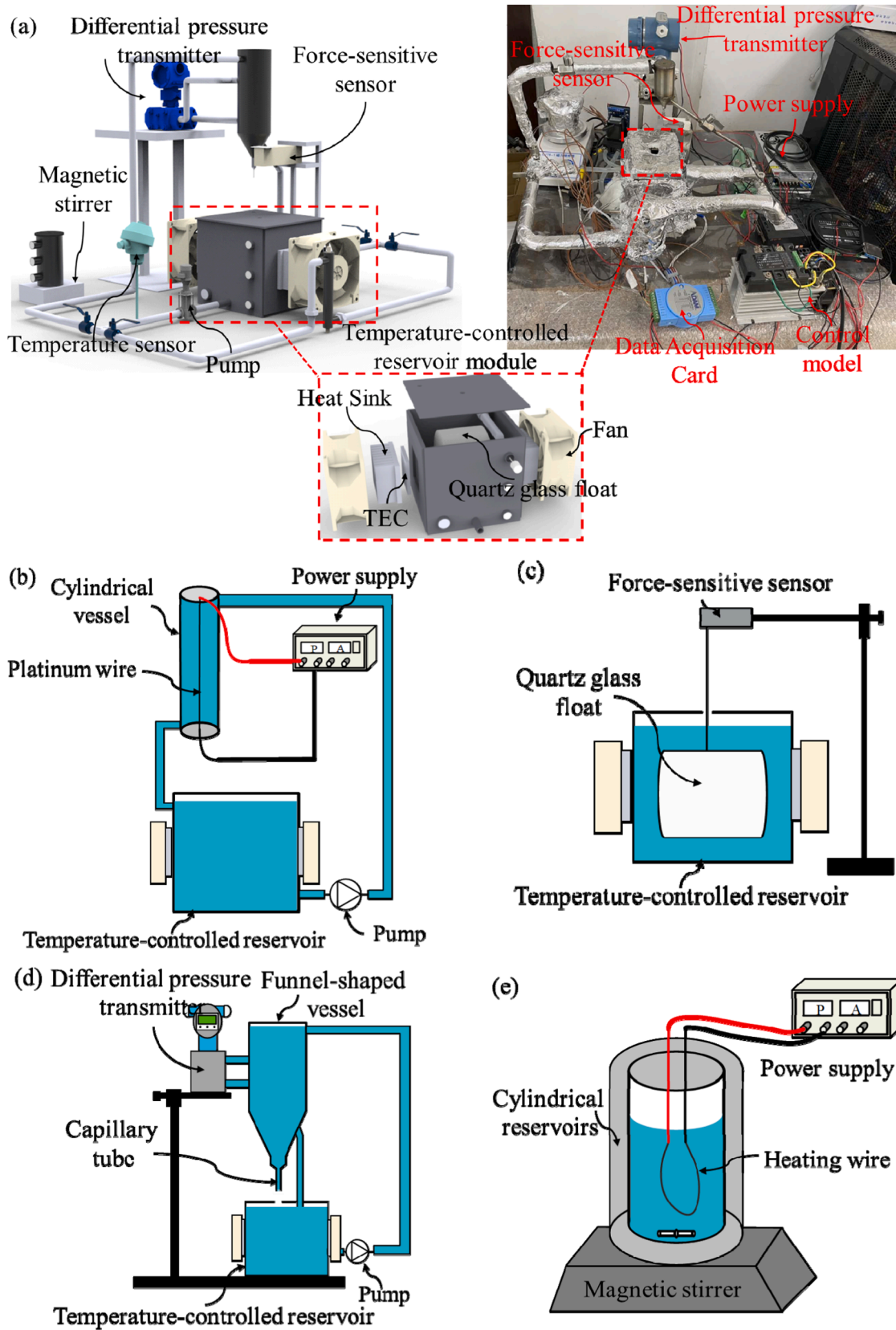


Fig. 1. Schematic of experimental setup: (a) Integrated experimental platform; (b) Thermal conductivity unit; (c) Density unit; (d) Dynamic viscosity unit; (e) Specific heat capacity unit.

controlled reservoir module is a stainless-steel reservoir with dimensions of 100 mm × 100 mm × 100 mm. Moreover, there are two thermoelectric cooling (TEC) modules with a refrigeration capacity of 60 W located on both sides of the reservoir, while two heating rods with a power of 150 W are arranged diagonally in the reservoir, thus enabling the temperature control of EFLs. In addition, three thermocouples are placed at 10 mm, 45 mm, and 80 mm from the reservoir bottom to detect the transient temperature characteristics of EFLs.

Fig. 1(b) illustrates the thermal conductivity measurement unit, including a cylindrical vessel, a peristaltic pump, a temperature-controlled reservoir, a platinum wire, and a direct current (DC) power supply. The cylindrical vessel is made of stainless steel with a radius of 5 mm and a height of 80 mm. A platinum wire with a radius of 0.025 mm and a length of 93 mm is arranged on its internal central axis, and the DC power supply adjusts its heat load. In addition, the ends of the platinum wire are equipped with a 1.5 mm diameter and 5 mm long PTFE insulation ring to reduce measurement errors.

Fig. 1(c) presents the density measurement unit, which consists of a quartz glass float, a force-sensitive sensor, and a temperature-controlled reservoir. The quartz glass float is a cylinder with a radius of 40 mm and a height of 50 mm. It is suspended from the force-sensitive sensor by a molybdenum wire and is submerged in the coolant, thus measuring the change in buoyancy.

Fig. 1(d) plots the Dynamic viscosity measurement unit, including a funnel-shaped vessel, a differential pressure transmitter, a temperature-controlled reservoir, and a peristaltic pump. The funnel-shaped vessel is fabricated from stainless steel with a radius of 20 mm and a height of 80 mm, and a capillary tube with a diameter of 0.85 mm is installed at its bottom to ensure that the EFL falls as droplets. The differential pressure transmitter is positioned on the side of the funnel-shaped vessel and has a differential pressure height of 30 mm.

Fig. 1(e) shows the specific heat capacity measurement unit, including a cylindrical reservoir, a magnetic stirrer, a heating wire, and a DC power supply. The stainless-steel cylindrical reservoir has a radius of 14.5 mm and a height of 60 mm. The DC power supply is used to adjust the electrical power of the heating wire. The magnetic stirrer is employed to ensure uniform heating of the EFL. In addition, three thermocouples are arranged at different locations in the cylindrical reservoir to monitor its temperature uniformity.

2.2. Measurement principles and methods

2.2.1. Thermal conductivity

Given the advantages of the transient hot-wire method in terms of its fast measurement speed, wide application range, and ability to avoid natural convection effects, this paper uses it to measure the thermal conductivity of liquids. During experimental measurements, a constant current (i.e., I) in a range of 50–100 mA is applied to the platinum wire and maintained for a short period ($\tau \leq 2 \sim 3$ s). Simultaneously, the voltages U_0 and U_1 across the platinum wire are recorded at intervals of 0.1 s before and after applying the current. Subject to all the assumptions of the transient hot-wire method, the thermal conductivity (i.e., λ) can be calculated by

$$\lambda = \frac{U_0^2 \kappa L \ln(\tau)}{4\pi L(U_1 - U_0)} + C \quad (1)$$

where κ and L are the temperature resistance coefficient ($\kappa=0.00392$) and effective length of a platinum wire, respectively, and C is a constant obtained by the least squares fit.

2.2.2. Density

From the perspective of simplicity and ease-operation, the hydrostatic suspension method is chosen to measure the expansion coefficient of liquids. It should be noted that this paper replaces the analytical balance with a force-sensitive sensor (range: 0–10 N; precision:

± 0.0001 N), which is more accurate and allows continuous measurements. During the measurement, the stable force-sensitive sensor readings (i.e., M_1 and M_2) are recorded when the glass float is submerged in the measured medium at temperatures T_1 and T_2 ($T_2 > T_1$). Subsequently, density at temperatures T_1 and T_2 can be calculated by

$$\rho(T_1) = \frac{M - M_1}{V_f g} \quad \rho(T_2) = \frac{M - M_2}{V_f g} \quad (2)$$

where M is the force-sensitive sensor reading when the glass float is not submerged in the measured medium, V_f is the glass float volume, and g denotes the gravitational acceleration.

2.2.3. Dynamic viscosity

This paper uses the capillary method [33] due to its simple structure, high applicability, and high measurement accuracy. To test working media with different viscosities, a removable stainless-steel needle with an inner diameter of 0.2–2 mm is used as a capillary herein. In the experiment, the differential pressure response of liquid-dropping processes for the calibration and measured medium is recorded by a differential pressure transmitter (precision: ± 1 Pa) at intervals of 0.5 s, and the average time (i.e., $\overline{\Delta\tau}$) for their differential pressure to change from the beginning to zero is calculated, respectively. Moreover, the EFL dynamic viscosity (i.e., μ) can be calculated by

$$\mu_f = \frac{\rho_f \overline{\Delta\tau_f}}{\rho_s \overline{\Delta\tau_s}} \mu_s \quad (3)$$

where ρ is density, subscripts f and s indicate measured and calibration coolants, respectively. In this paper, water with a known dynamic viscosity is selected as the calibration coolant for benchmark comparisons.

2.2.4. Specific heat capacity

This paper employs the widely available and highly accurate electrical power calorimetry method to measure the specific heat capacity of the measured medium. Furthermore, this paper uses a magnetic stirrer to accelerate the thermal stabilization process of the measured medium. Simultaneously, the measured medium is slowly heated with low power to a temperature near T and then steadily heated to T with constant power q , thereby minimizing ambient heat loss and improving measurement precision. During the measurement, the mass of the measured medium and cylindrical reservoir (i.e., m and m_c), the heating time (10–20 s) with constant power (i.e., $\Delta\tau$), and the corresponding temperature rise (i.e., ΔT and ΔT_c) are recorded at intervals of 0.1 s, then the specific heat capacity (i.e., c_p) of the measured medium can be calculated as

$$c_p = \frac{q\Delta\tau - c_{pc}m_c\Delta T_c}{m\Delta T} \quad (4)$$

2.3. Data reduction and uncertainty analysis

Uncertainties of directly measured parameters in experiments mainly arise from the inherent precision of measurement instruments. Generally, the relative uncertainty of a directly measured parameter (i.e., δ_d) can be calculated by

$$\delta_d = \frac{|\sigma_E|}{E_{\min}} \quad (5)$$

where σ_E is the precision of the measuring equipment for parameter E , and $E_{\min}$ denotes the minimum value of the parameter in this paper. In this experiment, the precisions of the related equipment for direct measurement parameters of temperature, mass, time, pressure, and current are ± 0.5 °C, ± 0.01 g, ± 0.1 s, ± 1 Pa and ± 0.001 mA, respectively. Therefore, the relative uncertainties for corresponding

parameters are 2.5%, 0.4%, 5%, 1.5% and 0.2%, respectively.

For indirectly measured parameters, their relative uncertainties (i.e., δ_i) in this paper is calculated based on the error transfer function proposed by Moffat [34], which is given by

$$\delta_i = \left[\left(\frac{\partial F}{\partial E_1} \right)^2 \sigma_{E_1}^2 + \left(\frac{\partial F}{\partial E_2} \right)^2 \sigma_{E_2}^2 + \dots + \left(\frac{\partial F}{\partial E_n} \right)^2 \sigma_{E_n}^2 \right]^{1/2} / F \quad (6)$$

where F represents a function relationship. Based on the above equation, the relative uncertainties of the parameters measured indirectly in experiments are calculated in Table 1.

3. Numerical methodology

3.1. Model description

To evaluate the EFL performance, this work investigates the pump-driven SPILC cabinet, which is promising for widespread applications in server cooling. As shown in Fig. 2(a), a simplified SPILC cabinet containing a single server motherboard is employed as the calculation area herein, with overall dimensions of 828 mm long, 522 mm wide, and 39 mm high. The structure schematic of the server motherboard is shown in Fig. 2(b), which includes the printed circuit board (PCB), central processing unit (CPU), graphics processing unit (GPU), dual inline memory module (DIMM), power supply unit (PSU), platform controller hub (PCH) and hard disk drive (HDD). The specific information for power devices is shown in Table 2.

Based on the above physical model, this paper constructs a three-dimensional steady-state model of the flow and thermal transport inside a SPILC cabinet. It should be noted that EFLs in this paper are incompressible fluids. Moreover, considering the high flow velocity of EFLs in practical engineering applications, the flow state herein is turbulent. In addition, the wall thickness of the SPILC cabinet and the environmental heat loss are ignored to simplify the calculation. According to the above assumptions and the three conservation laws, governing equations for a SPILC cabinet can be obtained as

Continuity equation

$$\frac{\partial(\rho \bar{u}_i)}{\partial x_i} = 0 \quad (7)$$

Momentum equation

$$\frac{\partial}{\partial x_j} (\rho \bar{u}_i \bar{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_i \partial x_j} + \rho \bar{g} + \frac{\partial}{\partial x_j} (-\rho \bar{u}_i \bar{u}_j') \quad (8)$$

Energy equations

$$\frac{\partial}{\partial x_i} (\rho c_p \bar{u}_i \bar{T}) = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} - \rho c_p \bar{u}_i \bar{T}' \right) \quad \text{for EFL} \quad (9)$$

$$\lambda \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} \right) = 0 \quad \text{for server motherboard} \quad (10)$$

where $\bar{T}$ is the average temperature, $\bar{p}$ is the average pressure, $\bar{u}_i$ denotes the velocity components on each axis (i.e., x_i), $\bar{u}_i'$ and $\bar{T}'$ denote the velocity and temperature pulsation value, respectively. Compared to other turbulence models (i.e., standard k - ϵ model and *Realizable* k - ϵ model), the *RNG* k - ϵ model [35] has a wide range of applicability, high

accuracy, and relatively low computational effort; thus, it is chosen to characterize the turbulent flow.

k -equation

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{\text{eff}} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \epsilon \quad (11)$$

ϵ -equation

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \epsilon \bar{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{\text{eff}} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k + G_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon \quad (12)$$

where k represents the turbulent kinetic energy, μ_{eff} denotes the effective kinetic viscosity coefficient, G_k and G_b denote the generation of turbulence kinetic energy due to the mean velocity gradients and buoyancy, respectively. In addition, R_ϵ is a correction factor for ϵ equation, which is given by

$$R_\epsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0) \epsilon^3}{1 + \gamma \eta^3} \frac{\epsilon^3}{k} \quad (13)$$

where η is the ratio of turbulence to mean strain time, and other constants are: $\alpha_k = 1.393$; $\alpha_\epsilon = 1.393$; $C_{1\epsilon} = 1.42$; $C_{2\epsilon} = 1.68$; $C_\mu = 0.0845$; $\eta_0 = 4.38$; $\gamma = 0.012$. In this paper, the initial temperature of the SPILC cabinet is 20 °C, the inlet EFL temperature is also 20 °C, and the volume flow rate of inlet EFL is 5.1 L/s. In addition, the walls of the SPILC cabinet are adiabatic boundary conditions, and the fluid-solid interface is considered to be slip-free and conjugate heat transfer.

3.2. Solution and validation

To investigate the advantages and disadvantages of different EFLs, this paper uses Fluent, a commercial software package in ANSYS 2021, to address the internal flow and heat transport characteristics of SPILC cabinets with different EFLs. Considering the complexity of server motherboards, this paper chose ICEM's unstructured grid to mesh the SPILC cabinet and refine the grid system for high-power devices (e.g., GPU, CPU, etc.). In order to couple the velocity and pressure variables during numerical solutions, the SIMPLE algorithm is employed in this paper. This method corrects for velocity by solving the pressure correction equation and carrying out several iterations until the continuity equation is satisfied. To discretize the control equations, the pressure correction equation is in the PRESTO! format, which is highly adaptable to high-rotation and rapidly changing pressure flows, and the momentum and energy equations are in the second-order format with high stability and accuracy, where the sub-relaxation coefficients for pressure, momentum, and energy are 0.2, 0.8, and 0.9, respectively. A high-precision solver is selected herein to improve precision and accuracy. The residual convergence criterion is 10^{-7} for the energy equation and 10^{-4} for other equations. Additionally, the CPU temperature and inlet pressure of the server motherboard are monitored separately. The calculation is considered to have converged when the residuals converge and monitoring targets no longer vary with iteration steps.

Considering that the computational grid affects the numerical results, six grid systems are selected in this paper to verify the grid independence. For a SPILC cabinet under the same operating conditions, Fig. 3 gives the CPU temperature rise and the inlet and outlet pressure losses for the SPILC cabinet under different grid systems. As shown in the figure, when the number of computing cells is greater than 4870,002, the variation in the CPU temperature rise and pressure loss is already small enough to exclude the effect of grid systems on the simulation results. Given this, the number of computing cells for the numerical calculations herein is at least 4870,002, considering the computational accuracy and consumption of computational resources.

To validate the reliability and accuracy of the current simulation, a numerical model based on the SPILC experiments of Luo et al. [23] is developed here for comparative study. Moreover, the relevant

Table 1
Relative uncertainties of indirectly measured parameters.

Parameter	Symbol	Unit	Relative uncertainty
Thermal conductivity	λ	W/(m·K)	11.12%
Dynamic viscosity	μ	mPa·s	3.42%
Density	ρ	kg/m ³	3.41%
Specific heat capacity	c_p	J/(kg·K)	5.01%

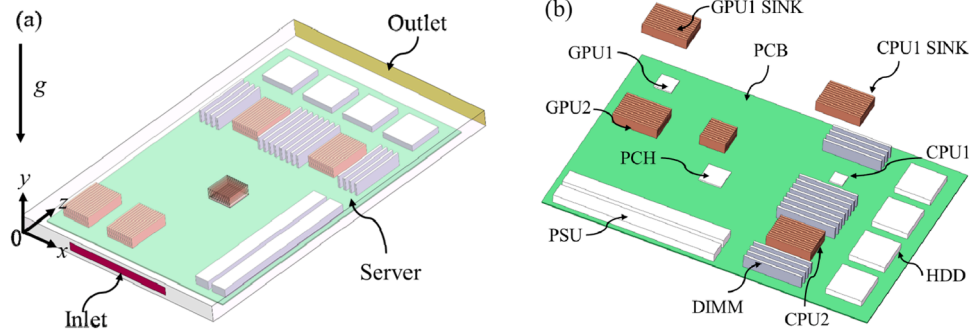


Fig. 2. Schematic diagram of the (a) SPILC cabinet and (b) server motherboard.

Table 2

Key parameters for power devices.

Type	Sizes (mm)	Thermal Design Power (W)	Number
CPU	30 × 30 × 3	270	2
GPU	40 × 40 × 3	170	2
DIMM	128 × 5 × 29	14.5	16
PSU	360 × 32 × 19	100	2
PCH	49 × 49 × 3	29.5	1
HDD	80 × 80 × 14.5	20	4

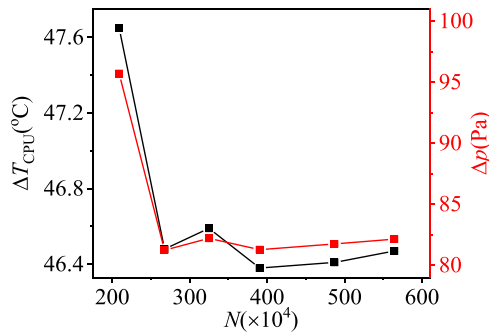


Fig. 3. Grid independence verification.

parameters of the numerical model are the same as those in the literature. A comparison of experimental data and numerical results for the heat source temperature in a SPILC system is plotted in Fig. 4(a). As it is indicated, the average temperature of the heat source simulated numerically herein fits well with the measurement data from the literature [23] with a maximum deviation of 7.25%. The reason for this deviation arose from the inevitable experimental errors during the experiment and the influence of relevant assumptions in the numerical

model. To further demonstrate the reliability of the simulation model, a comparison of the results of previous and current numerical simulations is given in Fig. 4(b). It is clear from the figure that there is a high degree of overlap between the two curves, which also confirms the accuracy of the simulation method in this work. The above analysis fully demonstrates that it is reasonable and reliable to study the performance of different EFLs based on the current numerical model.

4. Results and discussion

4.1. Validation of thermophysical property measurements

To verify the reliability of the constructed thermal property measurement platform, the thermal conductivity, kinetic viscosity, density, and specific heat capacity of standard media are measured. The measured and standard values of the thermal properties at different temperatures are given in Fig. 5. Notably, the measurement results for thermal conductivity, density, and specific heat capacity are validated using water as the standard media. However, for the measurement results of dynamic viscosity, anhydrous ethanol is used as the standard media. Among these, the average deviations for thermal conductivity, dynamic viscosity, density, and specific heat capacity are 3.68%, 4.44%, 0.81%, and 0.57%, respectively, which are all within the permissible limits of experimental error. Thus, the integrated experimental platform for thermal property measurement developed in this paper is reliable and trustworthy.

4.2. Evaluation of different immersion coolants

Given the complex and random nature of data center server architecture, the EFL selection for SPILC systems often requires significant computational resources and time consumption. To evaluate the thermal performance of EFLs quantitatively in advance, this paper derives a formula for calculating the FOM and compares the FOM of three

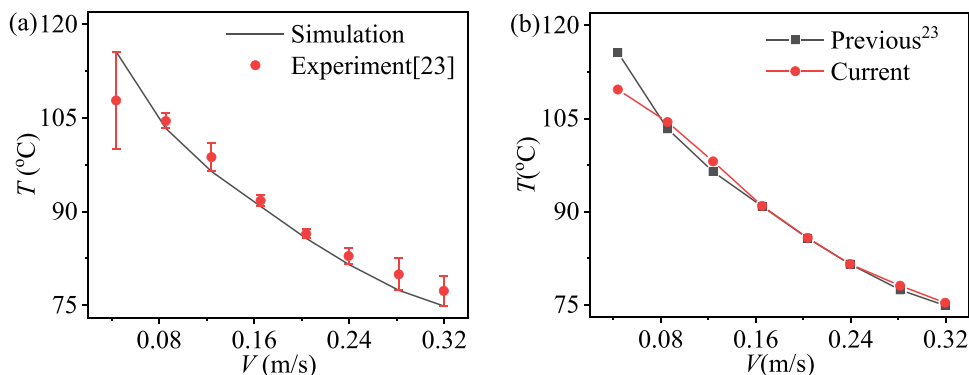


Fig. 4. Model validation: (a) Comparison of experiment and simulation; (b) Comparison of previous and current numerical results.

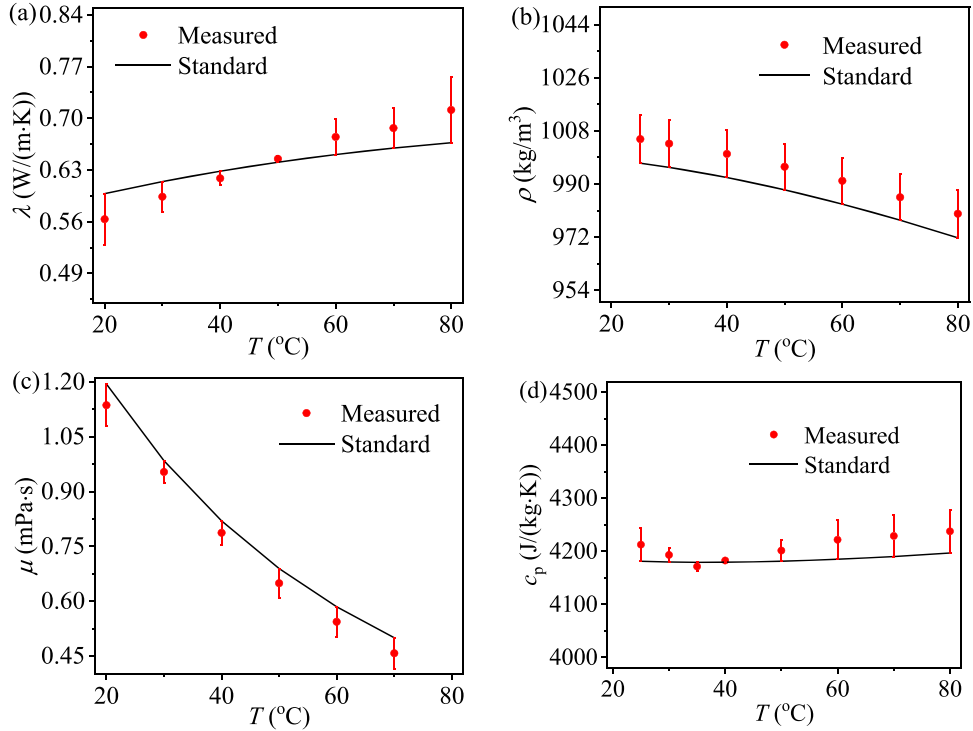


Fig. 5. Comparison of standard and measured thermophysical properties of cooling coolants (error bars stand for the absolute deviation): (a) Thermal conductivity, (b) Density, (c) Dynamic viscosity, and (d) Specific heat capacity.

different EFLs based on experimentally measured thermal properties for a pump-driven SPILC cabinet.

4.2.1. Definition of FOM

The operation of pump-driven SPILC cabinets is based on a convective heat transfer mechanism. However, there are several FOMs to compare the heat transfer performance of immersion coolants from their thermal properties. Considering that the flow state in pump-driven SPILC cabinets in real engineering is generally turbulent, the Mourtseff number is thus the most widely used FOM [36]. Moreover, as suggested by Vajjha and Das [37], this FOM can be obtained from the classical Dittus-Boelter equation [38] for base fluid, which is expressed as

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \quad (14)$$

where Nu denotes Nusselt number, Re represents Reynolds number, and Pr denotes Prandtl number. Considering that $Nu = hl/\lambda$, the convective heat transfer coefficient is given by

$$h = 0.023 \mu^{-0.4} \rho^{0.8} u^{0.8} l^{-0.2} c_p^{0.4} \lambda^{0.6} \quad (15)$$

where h is the convective heat transport coefficient, and l denotes characteristic length of the SPILC cabinet that equals hydraulic diameter. It is clear that the thermal performance of SPILC systems is only related to the flow characteristics, the EFL thermal properties, and the geometry of the SPILC cabinet. In other words, the FOM can be defined as

$$FOM = \mu^{-0.4} \rho^{0.8} c_p^{0.4} \lambda^{0.6} \quad (16)$$

According to the above equation, the FOM is positively correlated with density, specific heat capacity, and thermal conductivity, but inversely related to dynamic viscosity. In other words, the increase in density, specific heat capacity, and thermal conductivity are beneficial for SPILC thermal capacity, while high dynamic viscosity has a negative effect.

4.2.2. FOM for coolant candidates

This article selected three different EFLs from currently available commercial manufacturers for performance comparison, which are defined as EFL-F1, EFL-F2, and EFL-F3, respectively. Fig. 6 shows the thermal properties and FOM curves of the three EFLs measured on the experimental platform. From the figure, it is evident that the thermal properties of different EFLs do not vary significantly. Moreover, the thermal conductivity of different EFLs is relatively similar, whereas the dynamic viscosity exhibits several-fold differences. It is important to note that within the temperature range measured herein (20 °C–80 °C), EFL-F1 has the highest FOM, while EFL-F3 has the lowest FOM. Interestingly, the FOM for different EFLs exhibits a nearly monotonic linear increase with temperature. A careful analysis of the thermophysical property curves of different EFLs reveals that although the thermal conductivity and specific heat capacity of EFL-F3 are both larger than those of EFL-F1 and EFL-F2, and the density of EFL-F3 is close to that of EFL-F1, its dynamic viscosity is higher than that of EFL-F1 and EFL-F2. In other words, the increase in FOM caused by the high thermal conductivity and large specific heat capacity of EFL-F3 is weaker than the decrease in FOM caused by its high dynamic viscosity. This indicates that the dynamic viscosity of EFLs plays a more important role in the FOM than thermal conductivity and specific heat capacity. Therefore, the choice and design of EFLs should focus on their dynamic viscosity when there is no significant difference in their thermal conductivity.

From the above, it is clear that the dynamic viscosity of EFLs has a significantly higher degree of influence on the FOM than the other relevant thermal properties. To quantitatively analyze the importance of each relevant thermal property of EFLs, this paper obtains the weight factor of each relevant thermal property by taking partial derivatives of the FOM. From Eq. (16) above, the partial derivative of the FOM can be calculated as

$$\frac{\partial FOM}{\partial T} = \frac{\partial FOM}{\partial \mu} \frac{\partial \mu}{\partial T} + \frac{\partial FOM}{\partial \rho} \frac{\partial \rho}{\partial T} + \frac{\partial FOM}{\partial c_p} \frac{\partial c_p}{\partial T} + \frac{\partial FOM}{\partial \lambda} \frac{\partial \lambda}{\partial T} \quad (17)$$

Moreover, the relevant thermal properties of EFLs can be obtained as

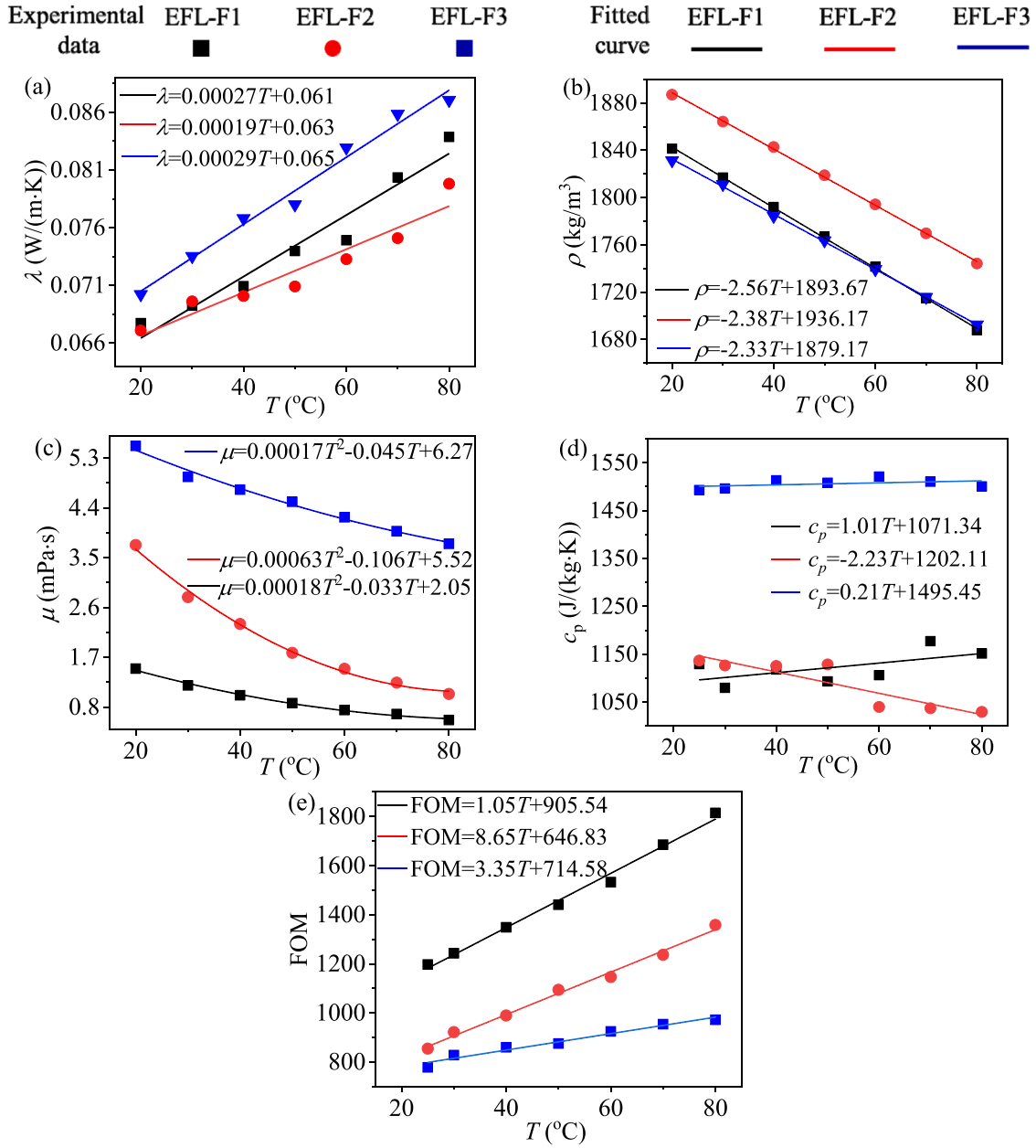


Fig. 6. Curves of (a) Thermal conductivity, (b) Density, (c) Dynamic viscosity, (d) Specific heat capacity, and (e) FOM with temperature for different EFLs.

a function of temperature using linear regression, and the above equation can then be further converted into

$$\begin{aligned} \partial FOM = & \left(\frac{\partial FOM}{\partial \mu} \frac{\partial \mu}{\partial T} \right)_{\mu} \partial T + \left(\frac{\partial FOM}{\partial \rho} \frac{\partial \rho}{\partial T} \right)_{\rho} \partial T + \left(\frac{\partial FOM}{\partial c_p} \frac{\partial c_p}{\partial T} \right)_{c_p} \partial T \\ & + \left(\frac{\partial FOM}{\partial \lambda} \frac{\partial \lambda}{\partial T} \right)_{\lambda} \partial T \end{aligned} \quad (18)$$

Finally, the weight factor (i.e., W_f) for each thermal property of EFLs can be obtained by normalizing the constants in percentage terms before ∂T in the above equation.

Fig. 7 shows the variation of weight factors of different thermophysical properties related to FOM with respect to temperature for different EFLs. As plotted in the figure, although the weight factors of various thermophysical properties of different EFLs are different, their overall trend of variation is basically the same. According to the importance of FOM, the ranking order of the thermophysical properties

is dynamic viscosity > thermal conductivity > density > specific heat capacity. It should be noted that with the increase in temperature, only the weight factor of dynamic viscosity increases, while the weight factors of thermal conductivity, density, and specific heat capacity all decrease. Interestingly, at low temperatures, the weight factor of thermal conductivity of EFL-F3 is slightly larger than that of dynamic viscosity. This phenomenon indicates that at low temperatures, where temperature gradients are small, the heat transfer of EFLs is primarily governed by thermal conduction. Therefore, a higher thermal conductivity can enhance the heat transfer capacity. However, as the temperature increases, the increase in thermal transport capacity caused by the high thermal conductivity is outweighed by the increase of thermal resistance caused by the thickening of the stagnant layer caused by the high dynamic viscosity, which weakens the heat transfer performance. Given this, the dominant role of dynamic viscosity and thermal conduction in influencing the thermal performance of EFLs is a game-like process, and it is highly sensitive to temperature variations. For most EFLs, there is no significant difference in the magnitude of their thermal

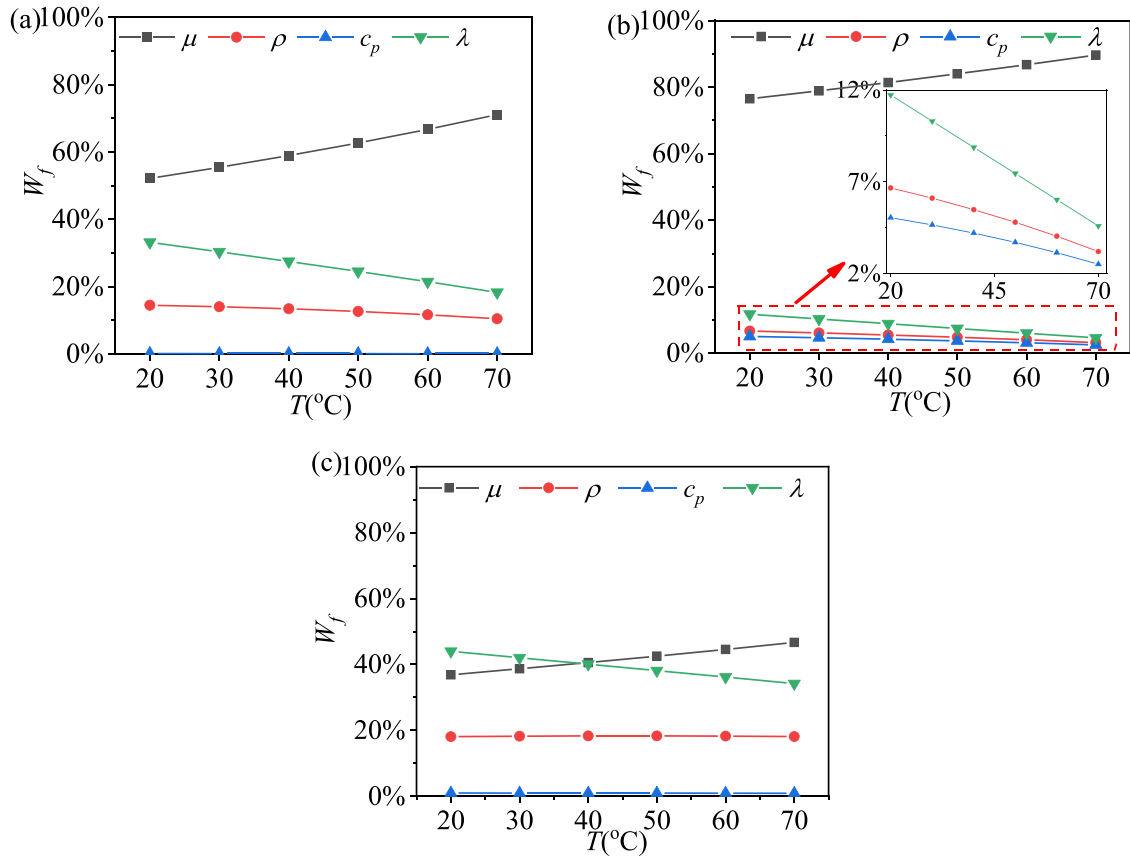


Fig. 7. Weight factors of thermophysical property for (a) EFL-F1, (b) EFL-F2, (c) EFL-F3.

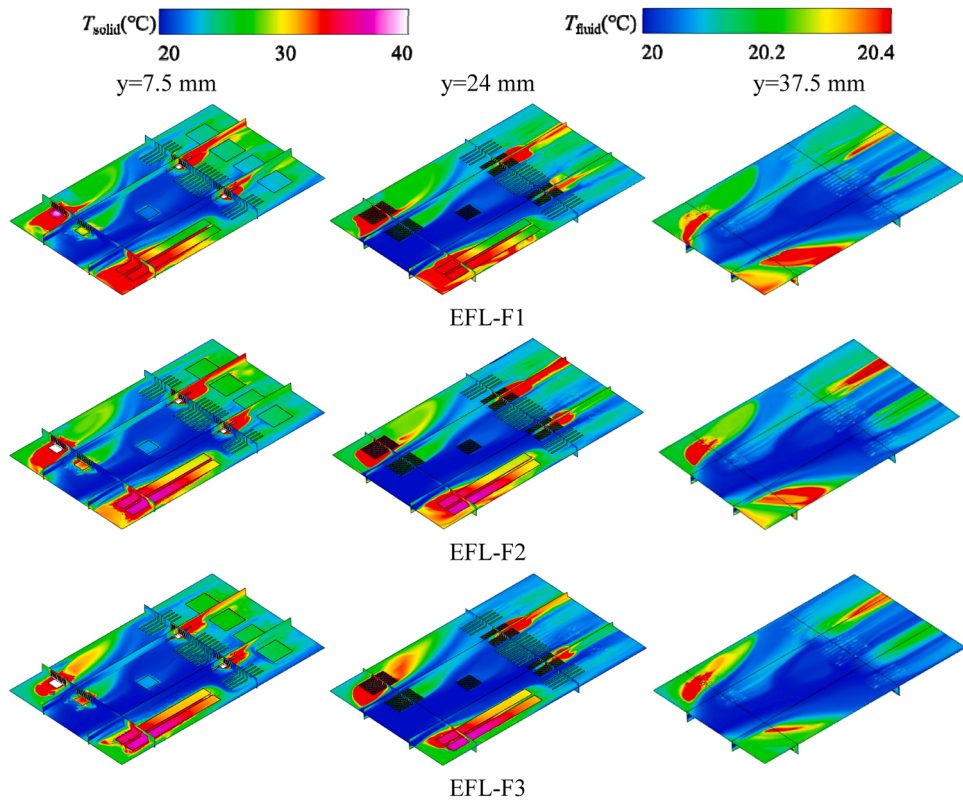


Fig. 8. Temperature field in the SPILC cabinet with different EFLs.

conductivity. Therefore, EFLs with lower dynamic viscosity are more likely to achieve better cooling performance for SPILC systems under relatively high temperatures.

4.3. Performance evaluation of SPILC systems with different EFLs

To verify the reliability of the FOM in evaluating the EFL performance, the SPILC cabinets with the three EFLs mentioned above are investigated numerically in this paper. In this context, some reference interfaces (i.e., $x = 156.5$ mm, 365.5 mm; $y = 7.5$ mm, 24 mm, 37.5 mm; $z = 135$ mm, 543 mm) are selected for an in-depth analysis of the thermal transport and flow characteristics of SPILC cabinets with different EFLs.

4.3.1. Temperature distribution characteristics

Fig. 8 presents a temperature distribution cloud of SPILC cabinets with different EFLs. It is clear from the figure that high power density electronics (e.g., CPU and GPU) are much hotter than the rest of the server. Moreover, it is clear that the electronics in the SPILC cabinet with EFL-F2 are at a higher temperature than in the SPILC cabinet with EFL-F1. As mentioned above, although the difference between the heat conductivity and specific heat capacity of EFL-F1 and EFL-F2 is not significant, the dynamic viscosity of EFL-F1 is smaller, resulting in a relatively thin stagnant layer on the electronic devices, which is conducive to heat dissipation of electronic devices on the server. Therefore, the average temperature of electronics in the SPILC cabinet with EFL-F1 is 9.3% lower than that in the SPILC cabinet with EFL-F2. This phenomenon is consistent with the findings of Choi et al. [39], who conducted experimental studies on the thermal performance of batteries using different immersion coolants. Their results also indicated that the highest temperature and temperature difference of batteries are relatively small when the hydrofluoroether (3 M Novec 7500) with low thermal conductivity and dynamic viscosity is used in the SPILC system. Additionally, it can be noted that although the electronics in the SPILC cabinet with EFL-F3 have a temperature closer to that of the SPILC cabinet with EFL-F2, the average temperature of EFL-F3 is slightly lower than that of EFL-F2. This phenomenon demonstrates that the cooling of electronics in the SPILC cabinet with EFL-F3 is less effective compared to EFL-F1 and EFL-F2. It is important to note that the specific heat capacity,

thermal conductivity, and dynamic viscosity of the EFL-F3 are higher than those of EFL-F1 and EFL-F2. However, although the larger specific heat capacity and thermal conductivity make for an enhanced thermal performance gain, the heat transport deterioration caused by the thickening of the stagnant layer due to the larger dynamic viscosity has a more significant impact. In summary, with the same inlet boundary and operating conditions, the EFL thermophysical properties have a significant impact on the SPILC systems. More importantly, the effect of EFL dynamic viscosity on the performance of SPILC cabinets is more prominent than specific heat capacity and heat conductivity, which is also in line with the findings above.

To quantify the impact of EFL type on the thermal transport performance of SPILC cabinets, the temperature rise of CPU1 and the temperature homogeneity coefficient of EFLs are given in Fig. 9. In particular, the temperature homogeneity coefficient is equal to the standard deviation (i.e., σ_T) of the temperature at 24 positions ($x = 140$ mm, 257 mm, 380 mm; $y = 15$ mm, 25 mm; $z = 195$ mm, 340 mm, 485 mm, 630 mm) inside the SPILC cabinet. Thus, the higher the temperature homogeneity coefficient, the worse the heat transfer uniformity of SPILC cabinets. It is clear from Fig. 9(a) that the CPU1 temperature rise is highest inside the SPILC cabinet with EFL-F3, while the CPU1 temperature rise corresponding to EFL-F1 is lowest, which is consistent with the previous analysis. According to statistics, the temperature rise of CPU1 inside SPILC cabinets with EFL-F1 decreased by 12.1% and 12.7%, respectively, compared to EFL-F2 and EFL-F3. This further demonstrates the superior heat transfer performance of EFL-F1. It is observed in Fig. 9 (b), the temperature homogeneity coefficients for three EFLs are all low due to the high inlet flow rate and the fully developed turbulent flow inside the SPILC cabinet. In addition, the temperature homogeneity coefficient inside the SPILC cabinets of EFL-F3 increased by 31.1% and 9.4% compared to EFL-F1 and EFL-F2, respectively. This is mainly because the dynamic viscosity of EFLs plays a dominant role in heat transfer performance. Therefore, even though EFL-F3 has a relatively higher thermal conductivity, its heat conduction benefits cannot compensate for the increased thermal resistance caused by the thickening of the boundary layer due to its high dynamic viscosity. Similarly, the average Nusselt number (as seen in Fig. 9(c)) in SPILC cabinets with EFL-F1 increased by 38.2% and 57.3%, respectively, compared to EFL-F2 and EFL-F3. Generally speaking, the dynamic viscosity of EFLs

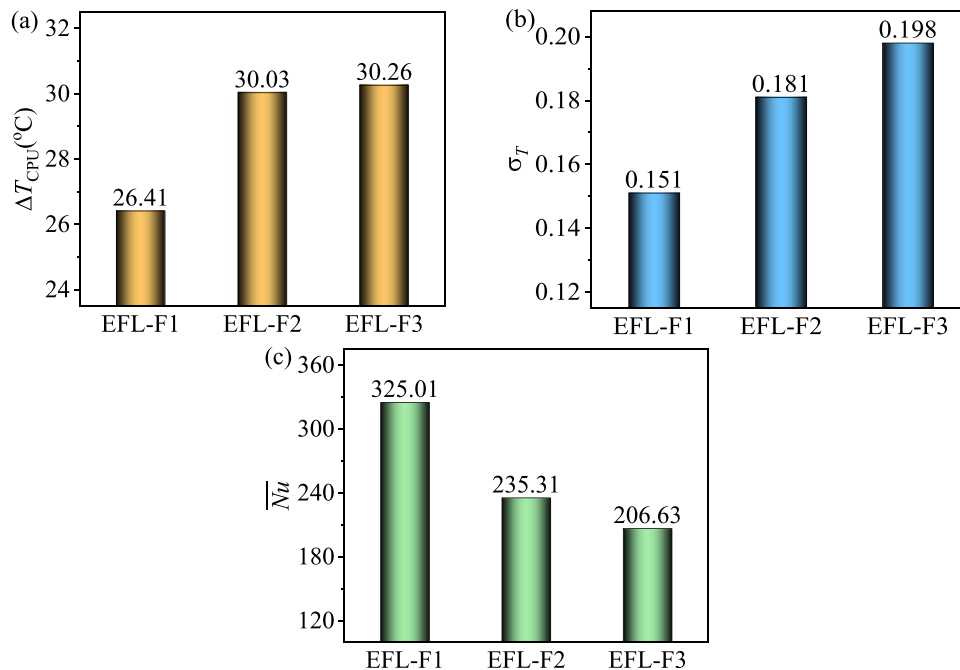


Fig. 9. Effect of EFL type on the thermal transport characteristics: (a) CPU1 temperature rise; (b) Temperature homogeneity coefficient; (c) Average Nusselt number.

plays a pivotal role in SPILC systems. The reduction in the dynamic viscosity enhances heat transfer and allows for a more uniform temperature distribution inside the SPILC cabinet.

4.3.2. Dynamic flow behaviors

To explore the flow characteristics in SPILC cabinets, the velocity distribution clouds inside the SPILC cabinets with different EFLs are given in Fig. 10. As is evident from the figure, since the asymmetrical arrangement of electronics on the server motherboard, the EFL flows at a higher rate in middle blank regions of the SPILC cabinet (where there are fewer electronics) and forms a tail flow area behind the critical electronics. Moreover, since the internal layout of the SPILC cabinets with different EFLs and the inlet boundaries are identical, the differences in the velocity fields are not significant between cases. Interestingly, it can still be noticed by closer inspection that the tail flow of EFL-F1 through the electronics on the server motherboard inside SPILC cabinets is shorter and smoother than that of EFL-F2 and EFL-F3. Besides, the average speed of EFLs above the server motherboard in the SPILC cabinet with EFL-F1 is higher than that of EFL-F2 and EFL-F3. The main reason for this phenomenon is the lower dynamic viscosity of the EFL-F1, which makes it easier to disengage after flowing through the electronics and has less interaction, which is one of the reasons why the thermal performance of the EFL-F1 is better than that of EFL-F2 and EFL-F3.

To quantitatively investigate the impact of different EFLs on internal flow characteristics inside SPILC cabinets, the pressure drop between the inlet and outlet of SPILC cabinets is introduced as an assessment criterion in this paper. As illustrated in Fig. 11, the pressure drop of the SPILC cabinet with EFL-F1 is reduced by 41.2% and 59%, respectively, compared to EFL-F2 and EFL-F3. This is mainly because the thermal conductivity of EFLs is generally small and not significantly different. Therefore, an increase in the dynamic viscosity of EFLs would result in a thicker stagnant layer formed on the surface of electronic components, leading to higher flow resistance losses within the SPILC system. Jithin

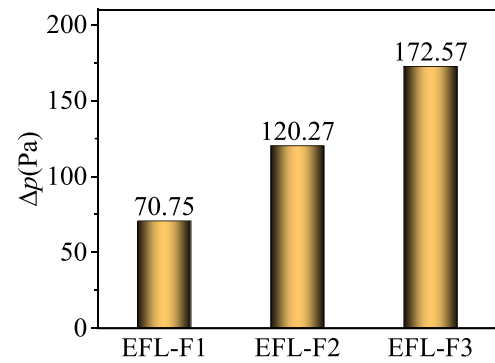


Fig. 11. Effect of EFL type on the pressure drop.

et al. [24] also found in their study on the thermal management of lithium batteries that although mineral oil and engineered fluids had similar thermal performance, the latter consumed 76.4% less power when operated at the same mass flow rate due to its lower dynamic viscosity. Given this, the dynamic viscosity dominates the influence of the flow and heat transport capacities of SPILC cabinets among the self-physical properties of EFLs.

5. Conclusion

In this paper, an integrated experimental platform was built to measure the thermophysical properties of EFLs for SPILC systems in data centers. The standard and measured thermophysical properties of water and anhydrous ethanol were compared to verify the accuracy and reliability of the experimental data. An evaluation indicator of FOM was proposed to characterize the thermal performance of EFLs in SPILC systems. Moreover, the importance of the thermal properties of EFLs to the FOM of SPILC systems was quantified. In addition, a numerical study

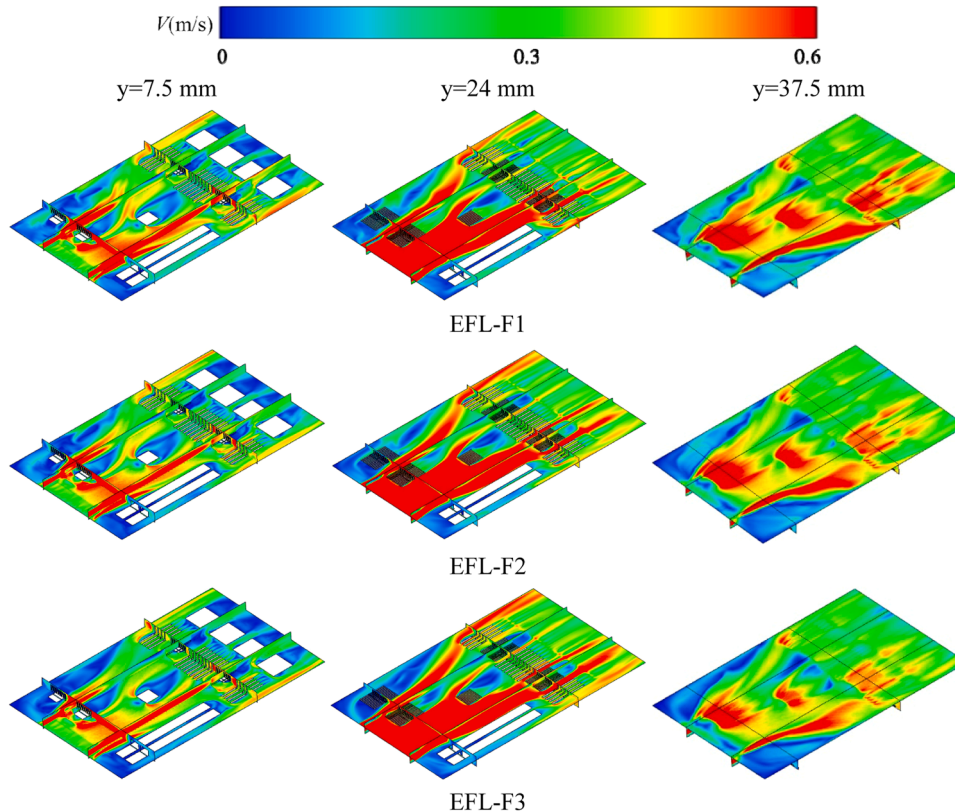


Fig. 10. Velocity field in the SPILC cabinet with different EFLs.

of flow and heat transport properties in SPILC systems with different EFLs was performed to demonstrate the generalizability of the FOM proposed herein. According to the above investigations, the following conclusions can be drawn.

- (1) According to the importance of FOM, the ranking order of the thermophysical properties for EFLs is dynamic viscosity > thermal conductivity > density > specific heat capacity. Moreover, with the increase in temperature, only the weight factor of dynamic viscosity increases, while weight factors of thermal conductivity, density, and specific heat capacity all decrease.
- (2) Compared to SPILC cabinets with EFL-F2 and EFL-F3, in the SPILC cabinet with EFL-F1, the CPU's maximum temperature decreases by 12.1%~12.7%, the temperature homogeneity coefficient reduces by 16.6% ~ 23.7%, and the pressure loss along the flow path decreases by 41.2% ~ 59%. Moreover, the average Nusselt number in SPILC cabinets with EFL-F1 increased by 38.2% and 57.3%, respectively, compared to EFL-F2 and EFL-F3.
- (3) For a pump-driven SPILC system, the thickening of the stagnant layer caused by the high dynamic viscosity of EFLs increases the thermal resistance and along-travel resistance loss. This results in a more significant loss in thermal performance compared to the heat transfer gain produced by increasing other thermophysical properties. Therefore, when selecting and designing EFL in data centers, efforts should be made to minimize their dynamic viscosity.

CRediT authorship contribution statement

Xi Chen: Numerical simulation, Formal analysis, Writing – original draft. **Yongping Huang:** Formal analysis, Writing – original draft. **Shijie Xu:** Experimental test, Data curation. **Chujin Bao:** Resources, Supervision. **Yangfan Zhong:** Validation. **Yongping Chen:** Methodology, Funding acquisition, Writing – review & editing. **Chengbin Zhang:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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